

```
param maxNode := 14;
```

```
param moduleCapacity := 10;
```

```
param: Links: link_nodeA link_nodeZ link_moduleCost :=
```

```
1 1 2 1
2 1 13 1
3 1 14 1
4 2 12 1
5 2 14 1
6 3 8 1
7 3 12 1
8 3 13 1
9 4 9 1
10 4 11 1
11 4 12 1
12 5 11 1
13 5 12 1
14 6 8 1
15 6 11 1
16 6 14 1
17 7 9 1
18 7 10 1
19 7 13 1
20 9 11 1
21 10 11 1
```

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;
```

```
param: Demands: demand_nodeA demand_nodeZ, demand_maxPath, demand_volume :=
```

```
1 1 2 4 45 #PaloAltoSanDiego
2 1 3 4 82 #PaloAltoBoulder
3 1 4 4 67 #PaloAltoWashington
4 1 5 4 90 #PaloAltoAtlanta
5 1 6 4 76 #PaloAltoUrbanaChampaign
6 1 7 4 66 #PaloAltoAnnArbor
7 1 8 4 61 #PaloAltoLincoln
8 1 9 4 68 #PaloAltoPrinceton
9 1 10 4 79 #PaloAltoIthaca
10 1 11 4 81 #PaloAltoPittsburgh
11 1 12 4 83 #PaloAltoHouston
12 1 13 4 42 #PaloAltoSaltLakeCity
13 1 14 4 61 #PaloAltoSeattle
14 2 3 4 64 #SanDiegoBoulder
15 2 4 4 71 #SanDiegoWashington
16 2 5 4 73 #SanDiegoAtlanta
17 2 6 4 64 #SanDiegoUrbanaChampaign
18 2 7 4 49 #SanDiegoAnnArbor
19 2 8 4 65 #SanDiegoLincoln
20 2 9 4 75 #SanDiegoPrinceton
21 2 10 4 72 #SanDiegoIthaca
22 2 11 4 62 #SanDiegoPittsburgh
23 2 12 4 68 #SanDiegoHouston
24 2 13 4 82 #SanDiegoSaltLakeCity
25 2 14 4 74 #SanDiegoSeattle
```

26 3 4 4 64 #BoulderWashington
27 3 5 4 84 #BoulderAtlanta
28 3 6 4 86 #BoulderUrbanaChampaign
29 3 7 4 51 #BoulderAnnArbor
30 3 8 4 62 #BoulderLincoln
31 3 9 4 75 #BoulderPrinceton
32 3 10 4 84 #BoulderIthaca
33 3 11 4 63 #BoulderPittsburgh
34 3 12 4 75 #BoulderHouston
35 3 13 4 85 #BoulderSaltLakeCity
36 3 14 4 62 #BoulderSeattle
37 4 5 4 75 #WashingtonAtlanta
38 4 6 4 63 #WashingtonUrbanaChampaign
39 4 7 4 89 #WashingtonAnnArbor
40 4 8 4 59 #WashingtonLincoln
41 4 9 4 67 #WashingtonPrinceton
42 4 10 4 86 #WashingtonIthaca
43 4 11 4 76 #WashingtonPittsburgh
44 4 12 4 83 #WashingtonHouston
45 4 13 4 62 #WashingtonSaltLakeCity
46 4 14 4 77 #WashingtonSeattle
47 5 6 4 63 #AtlantaUrbanaChampaign
48 5 7 4 69 #AtlantaAnnArbor
49 5 8 4 83 #AtlantaLincoln
50 5 9 4 65 #AtlantaPrinceton
51 5 10 4 72 #AtlantaIthaca
52 5 11 4 61 #AtlantaPittsburgh
53 5 12 4 74 #AtlantaHouston
54 5 13 4 85 #AtlantaSaltLakeCity
55 5 14 4 67 #AtlantaSeattle
56 6 7 4 72 #UrbanaChampaignAnnArbor
57 6 8 4 61 #UrbanaChampaignLincoln
58 6 9 4 61 #UrbanaChampaignPrinceton
59 6 10 4 60 #UrbanaChampaignIthaca
60 6 11 4 70 #UrbanaChampaignPittsburgh
61 6 12 4 82 #UrbanaChampaignHouston
62 6 13 4 75 #UrbanaChampaignSaltLakeCity
63 6 14 4 79 #UrbanaChampaignSeattle
64 7 8 4 68 #AnnArborLincoln
65 7 9 4 79 #AnnArborPrinceton
66 7 10 4 61 #AnnArborIthaca
67 7 11 4 68 #AnnArborPittsburgh
68 7 12 4 64 #AnnArborHouston
69 7 13 4 43 #AnnArborSaltLakeCity
70 7 14 4 44 #AnnArborSeattle
71 8 9 4 45 #LincolnPrinceton
72 8 10 4 49 #LincolnIthaca
73 8 11 4 48 #LincolnPittsburgh
74 8 12 4 76 #LincolnHouston
75 8 13 4 61 #LincolnSaltLakeCity
76 8 14 4 81 #LincolnSeattle
77 9 10 4 61 #PrincetonIthaca
78 9 11 4 61 #PrincetonPittsburgh
79 9 12 4 55 #PrincetonHouston

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80 9 13 4 57 #PrincetonSaltLakeCity
81 9 14 4 79 #PrincetonSeattle
82 10 11 4 61 #IthacaPittsburgh
83 10 12 4 46 #IthacaHouston
84 10 13 4 54 #IthacaSaltLakeCity
85 10 14 4 78 #IthacaSeattle
86 11 12 4 56 #PittsburghHouston
87 11 13 4 61 #PittsburghSaltLakeCity
88 11 14 4 87 #PittsburghSeattle
89 12 13 4 86 #HoustonSaltLakeCity
90 12 14 4 56 #HoustonSeattle
91 13 14 4 52 #SaltLakeCitySeattle
;

```

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set Demand_pathLinks[ 1, 1 ] := 1 ; #PaloAltoSanDiego
set Demand_pathLinks[ 1, 2 ] := 3 5 ;
set Demand_pathLinks[ 1, 3 ] := 2 8 7 4 ;
set Demand_pathLinks[ 1, 4 ] := 3 16 14 6 7 4;

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set Demand_pathLinks[ 2, 1 ] := 2 8 ; #PaloAltoBoulder
set Demand_pathLinks[ 2, 2 ] := 1 4 7 ;
set Demand_pathLinks[ 2, 3 ] := 3 16 14 6 ;
set Demand_pathLinks[ 2, 4 ] := 3 5 4 7 ;

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set Demand_pathLinks[ 3, 1 ] := 1 4 11 ; #PaloAltoWashington
set Demand_pathLinks[ 3, 2 ] := 2 8 7 11 ;
set Demand_pathLinks[ 3, 3 ] := 2 19 17 20 9 ;
set Demand_pathLinks[ 3, 4 ] := 1 4 13 12 20 9 ;

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set Demand_pathLinks[ 4, 1 ] := 1 4 13 ; #PaloAltoAtlanta
set Demand_pathLinks[ 4, 2 ] := 2 8 7 13;
set Demand_pathLinks[ 4, 3 ] := 3 5 4 13 ;
set Demand_pathLinks[ 4, 4 ] := 2 8 6 14 15 12 ;

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set Demand_pathLinks[ 5, 1 ] := 3 16 ; #PaloAltoUrbanaChampaign
set Demand_pathLinks[ 5, 2 ] := 1 5 16;
set Demand_pathLinks[ 5, 3 ] := 2 8 6 14;
set Demand_pathLinks[ 5, 4 ] := 1 4 7 6 14;

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set Demand_pathLinks[ 6, 1 ] := 2 19 ; #PaloAltoAnnArbor
set Demand_pathLinks[ 6, 2 ] := 1 4 7 8 19 ;
set Demand_pathLinks[ 6, 3 ] := 3 16 15 21 18 ;
set Demand_pathLinks[ 6, 4 ] := 2 8 6 14 15 21 18;

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set Demand_pathLinks[ 7, 1 ] := 2 8 6 ; #PaloAltoLincoln
set Demand_pathLinks[ 7, 2 ] := 3 16 14;
set Demand_pathLinks[ 7, 3 ] := 1 4 7 6 ;
set Demand_pathLinks[ 7, 4 ] := 1 5 16 14 ;

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set Demand_pathLinks[ 8, 1 ] := 2 19 17 ; #PaloAltoPrinceton
set Demand_pathLinks[ 8, 2 ] := 1 4 11 9 ;

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set Demand_pathLinks[ 8, 3 ] := 2 8 6 14 15 20 ;
set Demand_pathLinks[ 8, 4 ] := 3 16 15 20 ;

set Demand_pathLinks[ 9, 1 ] := 2 19 18 ;           #PaloAltoIthaca
set Demand_pathLinks[ 9, 2 ] := 3 16 15 21 ;
set Demand_pathLinks[ 9, 3 ] := 2 8 6 14 15 21 ;
set Demand_pathLinks[ 9, 4 ] := 1 4 13 12 ;

set Demand_pathLinks[ 10, 1 ] := 2 8 6 14 15 ;      #PaloAltoPittsburgh
set Demand_pathLinks[ 10, 2 ] := 2 19 18 21 ;
set Demand_pathLinks[ 10, 3 ] := 2 19 17 20 ;
set Demand_pathLinks[ 10, 4 ] := 3 16 15 ;

set Demand_pathLinks[ 11, 1 ] := 1 4 ;             #PaloAltoHouston
set Demand_pathLinks[ 11, 2 ] := 2 8 7 ;
set Demand_pathLinks[ 11, 3 ] := 3 5 4 ;
set Demand_pathLinks[ 11, 4 ] := 3 16 14 6 7 ;

set Demand_pathLinks[ 12, 1 ] := 2 ;               #PaloAltoSaltLakeCity
set Demand_pathLinks[ 12, 2 ] := 1 4 7 8 ;
set Demand_pathLinks[ 12, 3 ] := 3 5 4 7 8 ;
set Demand_pathLinks[ 12, 4 ] := 3 16 14 6 8 ;

set Demand_pathLinks[ 13, 1 ] := 3 ;               #PaloAltoSeattle
set Demand_pathLinks[ 13, 2 ] := 1 5 ;
set Demand_pathLinks[ 13, 3 ] := 2 8 6 14 16 ;
set Demand_pathLinks[ 13, 4 ] := 1 4 7 6 14 16 ;

set Demand_pathLinks[ 14, 1 ] := 1 2 8 ;           #SanDiegoBoulder
set Demand_pathLinks[ 14, 2 ] := 4 7 ;
set Demand_pathLinks[ 14, 3 ] := 5 16 14 6 ;
set Demand_pathLinks[ 14, 4 ] := 1 3 16 14 6 ;

set Demand_pathLinks[ 15, 1 ] := 4 11 ;           #SanDiegoWashington
set Demand_pathLinks[ 15, 2 ] := 1 2 8 7 11 ;
set Demand_pathLinks[ 15, 3 ] := 1 2 8 6 14 15 20 9 ;
set Demand_pathLinks[ 15, 4 ] := 4 13 12 20 9 ;

set Demand_pathLinks[ 16, 1 ] := 4 13 ;           #SanDiegoAtlanta
set Demand_pathLinks[ 16, 2 ] := 1 2 8 7 13 ;
set Demand_pathLinks[ 16, 3 ] := 1 2 8 6 14 15 12 ;
set Demand_pathLinks[ 16, 4 ] := 1 3 16 15 12 ;

set Demand_pathLinks[ 17, 1 ] := 4 7 6 14 ;       #SanDiegoUrbanaChampaign
set Demand_pathLinks[ 17, 2 ] := 1 2 8 6 14 ;
set Demand_pathLinks[ 17, 3 ] := 1 3 16 ;
set Demand_pathLinks[ 17, 4 ] := 4 13 12 15 ;

set Demand_pathLinks[ 18, 1 ] := 1 2 19 ;         #SanDiegoAnnArbor
set Demand_pathLinks[ 18, 2 ] := 4 11 9 17 ;
set Demand_pathLinks[ 18, 3 ] := 4 11 10 21 18 ;
set Demand_pathLinks[ 18, 4 ] := 5 16 15 21 18 ;

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set Demand_pathLinks[ 19, 1 ] := 1 2 8 6 ;           #SanDiegoLincoln
set Demand_pathLinks[ 19, 2 ] := 4 7 6 ;
set Demand_pathLinks[ 19, 3 ] := 5 16 14 ;
set Demand_pathLinks[ 19, 4 ] := 1 3 16 14 ;

set Demand_pathLinks[ 20, 1 ] := 4 11 9 ;           #SanDiegoPrinceton
set Demand_pathLinks[ 20, 2 ] := 5 16 15 20 ;
set Demand_pathLinks[ 20, 3 ] := 1 2 19 17 ;
set Demand_pathLinks[ 20, 4 ] := 1 3 16 15 20 ;

set Demand_pathLinks[ 21, 1 ] := 1 2 19 18 ;           #SanDiegoIthaca
set Demand_pathLinks[ 21, 2 ] := 5 16 15 21 ;
set Demand_pathLinks[ 21, 3 ] := 1 3 16 15 21 ;
set Demand_pathLinks[ 21, 4 ] := 4 13 12 ;

set Demand_pathLinks[ 22, 1 ] := 4 11 10 ;           #SanDiegoPittsburgh
set Demand_pathLinks[ 22, 2 ] := 5 16 15 ;
set Demand_pathLinks[ 22, 3 ] := 4 13 21 ;
set Demand_pathLinks[ 22, 4 ] := 4 11 9 20 ;

set Demand_pathLinks[ 23, 1 ] := 4 ;                 #SanDiegoHouston
set Demand_pathLinks[ 23, 2 ] := 1 2 8 7 ;
set Demand_pathLinks[ 23, 3 ] := 1 3 16 14 6 7 ;
set Demand_pathLinks[ 23, 4 ] := 5 16 14 6 7 ;

set Demand_pathLinks[ 24, 1 ] := 1 2 ;                 #SanDiegoSaltLakeCity
set Demand_pathLinks[ 24, 2 ] := 4 7 8 ;
set Demand_pathLinks[ 24, 3 ] := 5 16 14 6 8 ;
set Demand_pathLinks[ 24, 4 ] := 1 3 16 14 6 8 ;

set Demand_pathLinks[ 25, 1 ] := 5 ;                 #SanDiegoSeattle
set Demand_pathLinks[ 25, 2 ] := 1 3 ;
set Demand_pathLinks[ 25, 3 ] := 4 7 8 2 3 ;
set Demand_pathLinks[ 25, 4 ] := 1 2 8 6 14 16 ;

set Demand_pathLinks[ 26, 1 ] := 7 11 ;                 #BoulderWashington
set Demand_pathLinks[ 26, 2 ] := 6 14 15 10 ;
set Demand_pathLinks[ 26, 3 ] := 6 14 15 20 9 ;
set Demand_pathLinks[ 26, 4 ] := 8 19 17 9 ;

set Demand_pathLinks[ 27, 1 ] := 7 13 ;                 #BoulderAtlanta
set Demand_pathLinks[ 27, 2 ] := 6 14 15 21 12 ;
set Demand_pathLinks[ 27, 3 ] := 8 2 1 4 13 ;
set Demand_pathLinks[ 27, 4 ] := 8 19 18 12 ;

set Demand_pathLinks[ 28, 1 ] := 6 14 ;                 #BoulderUrbanaChampaign
set Demand_pathLinks[ 28, 2 ] := 8 19 18 21 15 ;
set Demand_pathLinks[ 28, 3 ] := 2 1 4 13 12 21 15 ;
set Demand_pathLinks[ 28, 4 ] := 8 2 3 16 ;

set Demand_pathLinks[ 29, 1 ] := 8 19 ;                 #BoulderAnnArbor
set Demand_pathLinks[ 29, 2 ] := 8 2 3 16 15 21 18 ;
set Demand_pathLinks[ 29, 3 ] := 7 11 9 17 ;
set Demand_pathLinks[ 29, 4 ] := 7 13 12 18 ;

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set Demand_pathLinks[ 30, 1 ] := 6 ;                                #BoulderLincoln
set Demand_pathLinks[ 30, 2 ] := 8 2 3 16 14 ;
set Demand_pathLinks[ 30, 3 ] := 7 13 12 21 15 14 ;
set Demand_pathLinks[ 30, 4 ] := 7 11 9 20 15 14 ;

set Demand_pathLinks[ 31, 1 ] := 6 14 15 20 ;
#BoulderPrinceton
set Demand_pathLinks[ 31, 2 ] := 7 11 9 ;
set Demand_pathLinks[ 31, 3 ] := 7 13 12 21 20 ;
set Demand_pathLinks[ 31, 4 ] := 8 19 17 ;

set Demand_pathLinks[ 32, 1 ] := 8 19 18 ;                                #BoulderIthaca
set Demand_pathLinks[ 32, 2 ] := 7 13 12 ;
set Demand_pathLinks[ 32, 3 ] := 6 14 15 21 ;
set Demand_pathLinks[ 32, 4 ] := 7 11 9 20 21 ;

set Demand_pathLinks[ 33, 1 ] := 6 14 15 ;
#BoulderPittsburgh
set Demand_pathLinks[ 33, 2 ] := 7 11 10 ;
set Demand_pathLinks[ 33, 3 ] := 7 13 12 21 ;
set Demand_pathLinks[ 33, 4 ] := 7 11 9 20 ;

set Demand_pathLinks[ 34, 1 ] := 7 ;                                #BoulderHouston
set Demand_pathLinks[ 34, 2 ] := 8 2 1 4 ;
set Demand_pathLinks[ 34, 3 ] := 6 14 15 10 11 ;
set Demand_pathLinks[ 34, 4 ] := 8 19 18 12 13 ;

set Demand_pathLinks[ 35, 1 ] := 8 ;                                #BoulderSaltLakeCity
set Demand_pathLinks[ 35, 2 ] := 7 4 1 2 ;
set Demand_pathLinks[ 35, 3 ] := 6 14 16 3 2 ;
set Demand_pathLinks[ 35, 4 ] := 6 14 15 21 18 19 ;

set Demand_pathLinks[ 36, 1 ] := 7 4 1 3 ;                                #BoulderSeattle
set Demand_pathLinks[ 36, 2 ] := 6 16 ;
set Demand_pathLinks[ 36, 3 ] := 8 2 3 ;
set Demand_pathLinks[ 36, 4 ] := 7 4 ;

set Demand_pathLinks[ 37, 1 ] := 11 13 ;
#WashingtonAtlanta
set Demand_pathLinks[ 37, 2 ] := 9 20 15 14 6 7 13 ;
set Demand_pathLinks[ 37, 3 ] := 10 21 12 ;
set Demand_pathLinks[ 37, 4 ] := 10 15 14 6 7 13 ;

set Demand_pathLinks[ 38, 1 ] := 10 15 ;
#WashingtonUrbanaChampaign
set Demand_pathLinks[ 38, 2 ] := 9 20 15 ;
set Demand_pathLinks[ 38, 3 ] := 11 7 6 14 ;
set Demand_pathLinks[ 38, 4 ] := 9 20 17 19 8 6 14 ;

set Demand_pathLinks[ 39, 1 ] := 9 17 ;                                #WashingtonAnnArbor
set Demand_pathLinks[ 39, 2 ] := 9 20 21 18 ;
set Demand_pathLinks[ 39, 3 ] := 11 7 8 19 ;
set Demand_pathLinks[ 39, 4 ] := 11 13 12 18 ;

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set Demand_pathLinks[ 40, 1 ] := 10 15 14 ;
#WashingtonLincoln
set Demand_pathLinks[ 40, 2 ] := 9 20 15 14 ;
set Demand_pathLinks[ 40, 3 ] := 11 7 6 ;
set Demand_pathLinks[ 40, 4 ] := 9 17 19 8 6 ;

set Demand_pathLinks[ 41, 1 ] := 9 ;                                #WashingtonPrinceton
set Demand_pathLinks[ 41, 2 ] := 10 20 ;
set Demand_pathLinks[ 41, 3 ] := 10 21 18 17 ;
set Demand_pathLinks[ 41, 4 ] := 11 13 12 21 20 ;

set Demand_pathLinks[ 42, 1 ] := 10 21 ;
#WashingtonIthaca
set Demand_pathLinks[ 42, 2 ] := 9 20 21 ;
set Demand_pathLinks[ 42, 3 ] := 9 17 18 ;
set Demand_pathLinks[ 42, 4 ] := 11 13 12 ;

set Demand_pathLinks[ 43, 1 ] := 10 ;                                #WashingtonPittsburgh
set Demand_pathLinks[ 43, 2 ] := 9 20 ;
set Demand_pathLinks[ 43, 3 ] := 11 13 12 21 ;
set Demand_pathLinks[ 43, 4 ] := 9 17 18 21 ;

set Demand_pathLinks[ 44, 1 ] := 11 ;                                #WashingtonHouston
set Demand_pathLinks[ 44, 2 ] := 9 20 21 12 13 ;
set Demand_pathLinks[ 44, 3 ] := 10 15 14 6 7 ;
set Demand_pathLinks[ 44, 4 ] := 9 17 19 8 7 ;

set Demand_pathLinks[ 45, 1 ] := 11 7 8 ;                                #WashingtonSaltLakeCity
set Demand_pathLinks[ 45, 2 ] := 10 15 14 6 8 ;
set Demand_pathLinks[ 45, 3 ] := 9 20 15 14 6 8 ;
set Demand_pathLinks[ 45, 4 ] := 9 17 19 ;

set Demand_pathLinks[ 46, 1 ] := 11 4 5 ;
#WashingtonSeattle
set Demand_pathLinks[ 46, 2 ] := 11 4 1 3 ;
set Demand_pathLinks[ 46, 3 ] := 9 17 19 2 3 ;
set Demand_pathLinks[ 46, 4 ] := 10 15 16 ;

set Demand_pathLinks[ 47, 1 ] := 12 21 15 ;
#AtlantaUrbanaChampaign
set Demand_pathLinks[ 47, 2 ] := 13 7 6 14 ;
set Demand_pathLinks[ 47, 3 ] := 12 18 17 20 15 ;
set Demand_pathLinks[ 47, 4 ] := 13 4 5 16 ;

set Demand_pathLinks[ 48, 1 ] := 12 18 ;                                #AtlantaAnnArbor
set Demand_pathLinks[ 48, 2 ] := 12 21 20 17 ;
set Demand_pathLinks[ 48, 3 ] := 13 7 8 19 ;
set Demand_pathLinks[ 48, 4 ] := 13 4 1 2 19 ;

set Demand_pathLinks[ 49, 1 ] := 13 7 6 ;                                #AtlantaLincoln

set Demand_pathLinks[ 49, 2 ] := 12 21 15 14 ;
set Demand_pathLinks[ 49, 3 ] := 12 18 19 8 6 ;

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set Demand_pathLinks[ 49, 4 ] := 13 4 5 16 14 ;

set Demand_pathLinks[ 50, 1 ] := 12 21 20 ;           #AtlantaPrinceton
set Demand_pathLinks[ 50, 2 ] := 12 18 17 ;
set Demand_pathLinks[ 50, 3 ] := 13 7 6 14 15 20 ;
set Demand_pathLinks[ 50, 4 ] := 13 11 9 ;

set Demand_pathLinks[ 51, 1 ] := 12 ;                 #AtlantaIthaca
set Demand_pathLinks[ 51, 2 ] := 13 11 10 21 ;
set Demand_pathLinks[ 51, 3 ] := 13 11 9 20 21 ;
set Demand_pathLinks[ 51, 4 ] := 13 11 9 17 18 ;

set Demand_pathLinks[ 52, 1 ] := 13 11 10 ;           #AtlantaPittsburgh
set Demand_pathLinks[ 52, 2 ] := 13 11 9 20 ;
set Demand_pathLinks[ 52, 3 ] := 13 7 6 14 15 ;
set Demand_pathLinks[ 52, 4 ] := 13 4 5 16 15 ;

set Demand_pathLinks[ 53, 1 ] := 13 ;                 #AtlantaHouston
set Demand_pathLinks[ 53, 2 ] := 12 21 10 11 ;
set Demand_pathLinks[ 53, 3 ] := 12 21 20 9 11 ;
set Demand_pathLinks[ 53, 4 ] := 12 18 17 9 11 ;

set Demand_pathLinks[ 54, 1 ] := 13 7 8 ;             #AtlantaSaltLakeCity
set Demand_pathLinks[ 54, 2 ] := 13 4 1 2 ;
set Demand_pathLinks[ 54, 3 ] := 12 18 19 ;
set Demand_pathLinks[ 54, 4 ] := 12 21 15 14 6 8 ;

set Demand_pathLinks[ 55, 1 ] := 13 4 1 3 ;           #AtlantaSeattle
set Demand_pathLinks[ 55, 2 ] := 13 4 5 ;
set Demand_pathLinks[ 55, 3 ] := 12 18 19 2 3 ;
set Demand_pathLinks[ 55, 4 ] := 12 21 15 16 ;

set Demand_pathLinks[ 56, 1 ] := 15 21 18 ;
#UrbanaChampaignAnnArbor
set Demand_pathLinks[ 56, 2 ] := 15 20 17 ;
set Demand_pathLinks[ 56, 3 ] := 15 10 9 17 ;
set Demand_pathLinks[ 56, 4 ] := 14 6 8 19 ;

set Demand_pathLinks[ 57, 1 ] := 14 ;                 #UrbanaChampaignLincoln
set Demand_pathLinks[ 57, 2 ] := 15 20 17 19 8 6 ;
set Demand_pathLinks[ 57, 3 ] := 15 10 11 7 6 ;
set Demand_pathLinks[ 57, 4 ] := 15 20 9 11 7 6 ;

set Demand_pathLinks[ 58, 1 ] := 15 20 ;
#UrbanaChampaignPrinceton
set Demand_pathLinks[ 58, 2 ] := 15 10 9 ;
set Demand_pathLinks[ 58, 3 ] := 15 21 18 17 ;
set Demand_pathLinks[ 58, 4 ] := 15 21 12 13 11 9 ;

set Demand_pathLinks[ 59, 1 ] := 15 21 ;
#UrbanaChampaignIthaca
set Demand_pathLinks[ 59, 2 ] := 15 20 17 18 ;
set Demand_pathLinks[ 59, 3 ] := 15 10 9 17 18 ;
set Demand_pathLinks[ 59, 4 ] := 15 10 9 20 21 ;

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set Demand_pathLinks[ 60, 1 ] := 15 ;                #UrbanaChampaignPittsburgh
set Demand_pathLinks[ 60, 2 ] := 14 6 7 11 10 ;
set Demand_pathLinks[ 60, 3 ] := 14 6 8 19 17 20 ;
set Demand_pathLinks[ 60, 4 ] := 14 6 7 13 12 21 ;

set Demand_pathLinks[ 61, 1 ] := 15 10 11 ;           #UrbanaChampaignHouston
set Demand_pathLinks[ 61, 2 ] := 15 20 9 11 ;
set Demand_pathLinks[ 61, 3 ] := 15 21 12 13 ;
set Demand_pathLinks[ 61, 4 ] := 14 6 7 ;

set Demand_pathLinks[ 62, 1 ] := 14 6 8 ;
#UrbanaChampaignSaltLakeCity
set Demand_pathLinks[ 62, 2 ] := 16 3 2 ;
set Demand_pathLinks[ 62, 3 ] := 15 21 18 19 ;
set Demand_pathLinks[ 62, 4 ] := 15 20 17 19 ;

set Demand_pathLinks[ 63, 1 ] := 16 ;                #UrbanaChampaignSeattle
set Demand_pathLinks[ 63, 2 ] := 14 6 8 2 3 ;
set Demand_pathLinks[ 63, 3 ] := 14 6 7 4 5 ;
set Demand_pathLinks[ 63, 4 ] := 14 6 7 4 1 3 ;

set Demand_pathLinks[ 64, 1 ] := 17 20 15 14 ;        #AnnArborLincoln
set Demand_pathLinks[ 64, 2 ] := 18 21 15 14 ;
set Demand_pathLinks[ 64, 3 ] := 19 8 6 ;
set Demand_pathLinks[ 64, 4 ] := 19 2 1 4 7 6 ;

set Demand_pathLinks[ 65, 1 ] := 17 ;                #AnnArborPrinceton
set Demand_pathLinks[ 65, 2 ] := 18 21 20 ;
set Demand_pathLinks[ 65, 3 ] := 18 21 10 9 ;
set Demand_pathLinks[ 65, 4 ] := 18 12 13 11 9 ;

set Demand_pathLinks[ 66, 1 ] := 18 ;                #AnnArborIthaca
set Demand_pathLinks[ 66, 2 ] := 17 20 21 ;
set Demand_pathLinks[ 66, 3 ] := 17 9 10 21 ;
set Demand_pathLinks[ 66, 4 ] := 17 9 11 13 12 ;

set Demand_pathLinks[ 67, 1 ] := 17 20 ;             #AnnArborPittsburgh
set Demand_pathLinks[ 67, 2 ] := 18 21 ;
set Demand_pathLinks[ 67, 3 ] := 17 9 10 ;
set Demand_pathLinks[ 67, 4 ] := 19 8 6 14 15 ;

set Demand_pathLinks[ 68, 1 ] := 18 12 13 ;          #AnnArborHouston
set Demand_pathLinks[ 68, 2 ] := 18 21 10 11 ;
set Demand_pathLinks[ 68, 3 ] := 17 9 11 ;
set Demand_pathLinks[ 68, 4 ] := 18 21 20 9 11 ;

set Demand_pathLinks[ 69, 1 ] := 19 ;                #AnnArborSaltLakeCity
set Demand_pathLinks[ 69, 2 ] := 17 20 15 14 6 8 ;
set Demand_pathLinks[ 69, 3 ] := 18 12 13 7 8 ;
set Demand_pathLinks[ 69, 4 ] := 18 21 10 11 7 8 ;

set Demand_pathLinks[ 70, 1 ] := 19 2 3 ;            #AnnArborSeattle
set Demand_pathLinks[ 70, 2 ] := 18 12 13 4 5 ;

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set Demand_pathLinks[ 70, 3 ] := 18 12 13 4 1 3 ;
set Demand_pathLinks[ 70, 4 ] := 17 20 15 16 ;

set Demand_pathLinks[ 71, 1 ] := 14 15 20 ;      #LincolnPrinceton
set Demand_pathLinks[ 71, 2 ] := 6 8 19 17 ;
set Demand_pathLinks[ 71, 3 ] := 14 15 10 9;
set Demand_pathLinks[ 71, 4 ] := 14 15 21 18 17 ;

set Demand_pathLinks[ 72, 1 ] := 6 7 13 12 ;      #LincolnIthaca
set Demand_pathLinks[ 72, 2 ] := 6 7 11 10 21 ;
set Demand_pathLinks[ 72, 3 ] := 6 7 11 9 20 21 ;
set Demand_pathLinks[ 72, 4 ] := 6 8 19 18 ;

set Demand_pathLinks[ 73, 1 ] := 14 15 ;          #LincolnPittsburgh
set Demand_pathLinks[ 73, 2 ] := 6 8 19 17 20 ;
set Demand_pathLinks[ 73, 3 ] := 6 8 19 18 21 ;
set Demand_pathLinks[ 73, 4 ] := 6 7 11 10 ;

set Demand_pathLinks[ 74, 1 ] := 6 7 ;            #LincolnHouston
set Demand_pathLinks[ 74, 2 ] := 14 15 10 11 ;
set Demand_pathLinks[ 74, 3 ] := 14 15 21 12 13 ;
set Demand_pathLinks[ 74, 4 ] := 14 15 20 9 11 ;

set Demand_pathLinks[ 75, 1 ] := 6 8 ;            #LincolnSaltLakeCity
set Demand_pathLinks[ 75, 2 ] := 14 16 3 2 ;
set Demand_pathLinks[ 75, 3 ] := 14 16 5 1 2 ;
set Demand_pathLinks[ 75, 4 ] := 6 7 4 1 2 ;

set Demand_pathLinks[ 76, 1 ] := 14 16 ;          #LincolnSeattle
set Demand_pathLinks[ 76, 2 ] := 6 7 4 5 ;
set Demand_pathLinks[ 76, 3 ] := 6 7 4 1 3 ;
set Demand_pathLinks[ 76, 4 ] := 6 8 2 3 ;

set Demand_pathLinks[ 77, 1 ] := 20 21 ;          #PrincetonIthaca
set Demand_pathLinks[ 77, 2 ] := 17 18 ;
set Demand_pathLinks[ 77, 3 ] := 9 10 21 ;
set Demand_pathLinks[ 77, 4 ] := 9 11 13 12 ;

set Demand_pathLinks[ 78, 1 ] := 20 ;             #PrincetonPittsburgh
set Demand_pathLinks[ 78, 2 ] := 9 10 ;
set Demand_pathLinks[ 78, 3 ] := 17 18 21 ;
set Demand_pathLinks[ 78, 4 ] := 9 11 13 12 21 ;

set Demand_pathLinks[ 79, 1 ] := 9 11 ;           #PrincetonHouston
set Demand_pathLinks[ 79, 2 ] := 20 21 12 13 ;
set Demand_pathLinks[ 79, 3 ] := 17 18 12 ;
set Demand_pathLinks[ 79, 4 ] := 17 18 21 10 11 ;

set Demand_pathLinks[ 80, 1 ] := 17 19 ;          #PrincetonSaltLakeCity
set Demand_pathLinks[ 80, 2 ] := 20 15 14 6 8 ;
set Demand_pathLinks[ 80, 3 ] := 9 11 7 8 ;
set Demand_pathLinks[ 80, 4 ] := 20 21 18 19 ;

set Demand_pathLinks[ 81, 1 ] := 20 15 16 ;      #PrincetonSeattle

```

```

set Demand_pathLinks[ 81, 2 ] := 17 19 2 3 ;
set Demand_pathLinks[ 81, 3 ] := 9 11 4 5 ;
set Demand_pathLinks[ 81, 4 ] := 9 11 4 1 3 ;

set Demand_pathLinks[ 82, 1 ] := 21 ;      #IthacaPittsburgh
set Demand_pathLinks[ 82, 2 ] := 18 17 20 ;
set Demand_pathLinks[ 82, 3 ] := 18 17 9 10 ;
set Demand_pathLinks[ 82, 4 ] := 12 13 11 10 ;

set Demand_pathLinks[ 83, 1 ] := 12 13 ;      #IthacaHouston
set Demand_pathLinks[ 83, 2 ] := 21 10 11 ;
set Demand_pathLinks[ 83, 3 ] := 21 20 9 11 ;
set Demand_pathLinks[ 83, 4 ] := 18 17 10 11 ;

set Demand_pathLinks[ 84, 1 ] := 18 19 ;      #IthacaSaltLakeCity
set Demand_pathLinks[ 84, 2 ] := 21 15 1 6 8 ;
set Demand_pathLinks[ 84, 3 ] := 12 13 7 8 ;
set Demand_pathLinks[ 84, 4 ] := 21 10 11 7 8 ;

set Demand_pathLinks[ 85, 1 ] := 18 19 2 3 ;      #IthacaSeattle
set Demand_pathLinks[ 85, 2 ] := 21 15 16 ;
set Demand_pathLinks[ 85, 3 ] := 12 13 4 5 ;
set Demand_pathLinks[ 85, 4 ] := 12 13 4 1 3 ;

set Demand_pathLinks[ 86, 1 ] := 10 11 ;      #PittsburghHouston
set Demand_pathLinks[ 86, 2 ] := 21 12 13 ;
set Demand_pathLinks[ 86, 3 ] := 20 9 11 ;
set Demand_pathLinks[ 86, 4 ] := 15 14 6 7 ;

set Demand_pathLinks[ 87, 1 ] := 20 17 19 ;      #PittsburghSaltLakeCity
set Demand_pathLinks[ 87, 2 ] := 15 14 6 8 ;
set Demand_pathLinks[ 87, 3 ] := 10 11 7 8 ;
set Demand_pathLinks[ 87, 4 ] := 21 12 13 7 8 ;

set Demand_pathLinks[ 88, 1 ] := 20 17 19 2 3 ;      #PittsburghSeattle
set Demand_pathLinks[ 88, 2 ] := 15 16 ;
set Demand_pathLinks[ 88, 3 ] := 10 11 4 5 ;
set Demand_pathLinks[ 88, 4 ] := 10 11 4 1 3 ;

set Demand_pathLinks[ 89, 1 ] := 7 8 ;      #HoustonSaltLakeCity
set Demand_pathLinks[ 89, 2 ] := 4 1 2 ;
set Demand_pathLinks[ 89, 3 ] := 13 12 18 19 ;
set Demand_pathLinks[ 89, 4 ] := 11 10 15 14 6 8 ;

set Demand_pathLinks[ 90, 1 ] := 4 5 ;      #HoustonSeattle
set Demand_pathLinks[ 90, 2 ] := 4 1 3 ;
set Demand_pathLinks[ 90, 3 ] := 7 6 14 16 ;
set Demand_pathLinks[ 90, 4 ] := 7 8 2 3 ;

set Demand_pathLinks[ 91, 1 ] := 2 3 ;      #SaltLakeCitySeattle
set Demand_pathLinks[ 91, 2 ] := 8 6 14 16 ;
set Demand_pathLinks[ 91, 3 ] := 8 7 4 5 ;
set Demand_pathLinks[ 91, 4 ] := 8 7 4 1 3 ;

```

```
param y :=  
1 56  
2 75  
3 41  
4 82  
5 33  
6 75  
7 78  
8 84  
9 46  
10 58  
11 80  
12 43  
13 92  
14 63  
15 101  
16 81  
17 69  
18 52  
19 101  
20 40  
21 41  
;
```